N100 FINANCIAL INTELLIGENCE

Comprehensive Financial Analyst Guide & System Architecture

Production Reference Manual for Data Pipelines, Screener Engines, Peer Radar Analytics, NLP Heuristics, KMeans Archetypes, and REST API Services across the Nifty 100 Universe.

Project:	N100 Financial Intelligence Platform
Release:	Sprint 6 Final Production Release (Sprint Sign-Off)
Target Index:	Nifty 100 Index (92 Operational Companies)
Coverage Period:	FY2011 – FY2024 (14 Fiscal Years)
Backend Tech:	Python 3.14 / SQLite 3 / FastAPI / Uvicorn / Scikit-Learn
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Executive Summary & Table of Contents

The N100 Financial Intelligence Platform is an institutional-grade financial data engineering and analytics system built for the Nifty 100 universe. Over six successive development sprints, the platform has evolved from raw heterogeneous Excel spreadsheets into an audited, deterministic, and highly performant analytics foundation. This guide documents the end-to-end methodologies, mathematical formulations, architectural decisions, and operational runbooks for professional equity research analysts, quantitative developers, and systems operators.

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Key Sprint 6 Delivery Milestones

- **16+ Production REST Endpoints:** High-throughput FastAPI backend serving real-time company financials, screener results, peer radar percentiles, and PDF downloads.
- **KMeans Archetype Profiling:** K=5 clustering model assigning all 92 companies with 10 validated financial features and diagnostic elbow/correlation plots.
- **Complete Automated Test Suite:** Over 235 automated unit, integration, and performance tests with zero failures verified in automated HTML reports.

Chapter 1: System Architecture & Ingestion Pipeline

The platform is architected around a strict decoupled unidirectional data pipeline. Raw data sources from capital market vendors and filing portals undergo normalization, integrity validation, metric derivation, and machine learning inference before serving analytical clients.

Core Architecture Tiers

Tier	Component	Key Responsibilities
Ingestion & Raw	12 Excel Master Files	Raw P&L, BS, CF, Ratios, Prices, Market Cap, Sectors, Documents.
ETL & Cleansing	src/etl/normaliser.py	Date normalization, fiscal year extraction, ticker standardization, type coercion.
Storage Engine	nifty100.db (SQLite 3)	Indexed relational schema with foreign key enforcement and write-ahead logging.
Analytics Core	src/analytics/	Ratio engine, CAGR turnaround detection, cashflow intelligence, peer percentiles.
Machine Learning	src/analytics/clustering.py	Median imputation, StandardScaler, KMeans (K=5), Z-score outlier detection.
REST API Tier	src/api/ (FastAPI)	16+ REST endpoints, CORS middleware, request timing logging, OpenAPI docs.
Presentation	Streamlit & ReportLab	9 interactive Streamlit screens and batch PDF tearsheet generators.

Database Schema Design & Normalization

The SQLite database contains 14 normalized tables. Companies form the root entity indexed by integer company_id (1 to 92). All child financial tables (profitandloss, balancesheet, cashflow, financial_ratios) maintain composite uniqueness over (company_id, year).

- **Idempotent Optimization:** 7 covering B-Tree indexes (e.g., idx_finratios_year, idx_pl_company_year) accelerate queries by 40–60%, transforming hot table scans into indexed B-Tree searches.
- **Foreign Key Enforcement:** SQLite PRAGMA foreign_keys = ON is enforced on every connection, preventing orphan child records.

Chapter 2: Data Quality & Master Validation Framework

Data integrity is guaranteed through the automated Data Quality (DQ) framework containing 16 formal validation rules. Rather than coercing dirty data silently to zeros or dropping anomalous rows without audit trails, the platform flags, categorizes, and logs every anomaly.

Rule ID	Rule Description	Tolerance / Condition	Status
DQ-01	Company Primary Key Uniqueness	100% Unique, Non-null, strictly positive	PASSED
DQ-02	P&L Composite Key (Company + Year)	Zero duplicates across all 1,073 records	PASSED
DQ-03	Balance Sheet Composite Key	Zero duplicates across all 1,058 records	PASSED
DQ-04	Cash Flow Composite Key	Zero duplicates across all 1,064 records	PASSED
DQ-05	Financial Ratios Composite Key	Zero duplicates across all 1,288 records	PASSED
DQ-06	Foreign Key Integrity (Ratios -> Companies)	Zero orphan child records in financial_ratios	PASSED
DQ-07	Foreign Key Integrity (P&L -> Companies)	Zero orphan child records in profitandloss	PASSED
DQ-08	Foreign Key Integrity (BS -> Companies)	Zero orphan child records in balancesheet	PASSED
DQ-09	Foreign Key Integrity (CF -> Companies)	Zero orphan child records in cashflow	PASSED
DQ-10	Balance Sheet Total Asset Positivity	> 99.9% positive; 1 historical warning (Adani Energy 2014)	PASSED (Flagged)
DQ-11	Balance Sheet Equation Sanity	Total Assets ≈ Total Liabilities + Total Equity	PASSED
DQ-12	Ticker Symbol Format Consistency	Non-empty, uppercase alphanumeric symbols	PASSED
DQ-13	ISIN Code Compliance	12-character format starting with 'IN' prefix	PASSED
DQ-14	Peer Group Referential Integrity	Zero invalid peer links	PASSED

Handling of Missing and TTM Observations

- **Trailing Twelve Months (TTM):** TTM data points from source files are explicitly decoupled from fiscal-year sequences and never assigned artificial year integers to avoid biasing multi-year time-series analysis.
- **Zero vs NULL Semantics:** NULL values are never coerced to zero. In financial ratios, a NULL indicates missing or inapplicable data (e.g., bank debt-to-equity), whereas zero denotes genuine absence of value (e.g., zero debt).

Chapter 3: Financial Ratio Engine & KPI Formulations

The financial ratio engine (src/analytics/ratios.py) derives 50+ Key Performance Indicators spanning profitability, leverage, coverage, efficiency, and growth. Crucially, the engine handles mathematical edge cases deterministically without raising runtime exceptions.

Metric Name	Formula / Derivation	Edge Case & Mathematical Handling
Net Profit Margin (%)	(Net Profit / Sales) * 100	Returns None if Sales <= 0 or missing; handles negative net profit.
Operating Margin (%)	(Operating Profit / Sales) * 100	Validates against source OPM; logs discrepancy if divergence > 2%.
Return on Equity (%)	(Net Profit / Net Worth) * 100	Returns None if Net Worth <= 0 (negative equity creates misleading high
Return on Assets (%)	(Net Profit / Total Assets) * 100	Returns None if Total Assets <= 0; tracks asset utilization efficiency.
Debt-to-Equity	Borrowings / (Equity + Reserves)	Returns 0.0 if Borrowings == 0; Financials sector automatically exempted.
Interest Coverage	EBIT / Interest Expense	Returns None / 'Debt Free' if Interest == 0; flags warning if ICR < 1.5x.
Asset Turnover	Sales / Total Assets	Measures revenue efficiency per rupee of assets; returns None if assets =
Free Cash Flow	Operating CF + Investing CF	Computed directly from cashflow statements in crores.
5-Year CAGR	((End / Start) ^ (1/n) - 1) * 100	Turnaround detection: Negative to Positive flagged as 'TURNAROUND'.
Composite Quality	0–100 Weighted Score	Winsorized P10/P90 percentile composite across profitability and leverage

Compound Annual Growth Rate (CAGR) Turnaround Logic

Standard CAGR formulas fail when the base year is negative or zero (e.g., turnaround from a net loss to profit). The engine classifies each CAGR observation into one of six audited states: NORMAL, TURNAROUND, DECLINE_TO_LOSS, BOTH_NEGATIVE, ZERO_BASE, or INSUFFICIENT.

Chapter 4: Config-Driven Screener Engine & Presets

The screener engine (src/screener/engine.py) implements declarative stock screening parameterized via config/screener_config.yaml. Operating strictly over the latest observation year (FY2024), it executes multi-metric Boolean conjunctions (AND logic) while enforcing business exceptions.

Preset Name	Core Filter Criteria	Passing Count (FY24)
Quality Compounder	ROE ≥ 15%, NPM ≥ 10%, D/E ≤ 1.0 (Exc. Fin), 5Y Rev CAGR ≥ 8%	34 Companies
Value Pick	P/E ≤ 35, P/B ≤ 8.0, ROE ≥ 12%, D/E ≤ 1.5 (Exc. Fin)	10 Companies
Growth Accelerator	5Y Rev CAGR ≥ 12%, 5Y PAT CAGR ≥ 15%, 5Y EPS CAGR ≥ 12%, ROE ≥ 12%	25 Companies
Dividend Champion	Dividend Yield ≥ 2.0%, Payout Ratio 20–150%, ROE ≥ 10%, D/E ≤ 1.5	19 Companies
Debt-Free Blue Chip	D/E ≤ 0.1 (Strict, no exemption), ROE ≥ 12%, NPM ≥ 8%, ICR ≥ 10x	21 Companies
Turnaround Watch	5Y Rev CAGR ≥ 8%, 5Y PAT CAGR ≥ 15%, 5Y EPS CAGR ≥ 12%, ROE ≥ 8%	37 Companies

Special Financial Sector & Debt-Free Rule Handling

- **Financials D/E Exemption:** Banks and NBFCs operate with high regulatory leverage by design (deposits as liabilities). The screener engine checks sector classification and automatically waives D/E ceilings for companies in Financials, Banking, and Insurance.
- **Strict Opt-Out:** In strategies specifically targeting unleveraged balance sheets (such as debt_free_blue_chip), the exemption is disabled via apply_exception: false, requiring all passing companies to meet strict leverage caps.
- **Deterministic Boundary Validation:** Acceptance criteria mandate that every preset returns between 5 and 50 companies on the universe, preventing empty results or unselective universes.

Chapter 5: Peer Group Analytics & Radar Visualizations

To contextualize corporate performance against direct industry competitors, the platform organizes companies into 11 specialized peer groups and computes intra-group percentile rankings across 10 vital KPIs.

11 Institutional Peer Groups

Peer Group	Representative Companies	Benchmark Characteristics
IT Services & Tech	TCS, INFY, HCLTECH, TECHM, LTIM	Asset-light, high ROE (>25%), negative net debt, export-driven.
Private Sector Banks	HDFCBANK, ICICIBANK, KOTAKBANK, AXISBANK	High NIM (>3.5%), low NPA ratios, exempt from D/E filtering.
Public Sector Banks	SBIN, BANKBARODA, PNB, CANBK	High asset base, sovereign backing, lower valuation multiples.
FMCG & Consumer Goods	HINDUNILVR, ITC, NESTLEIND, BRITANNIA	High operating margin, strong pricing power, high dividend payout.
Automobile OEMs	MARUTI, TATAMOTORS, M&M, BAJAJ-AUTO	Cyclical CapEx, EV transition exposure, moderate leverage.
Oil, Gas & Energy	RELIANCE, ONGC, BPCL, IOC, GAIL	High capital intensity, commodity price sensitivity, high cash flows.
Pharmaceuticals	SUNPHARMA, DRREDDY, CIPLA, DIVISLAB	High R&D expenditure, global regulatory scrutiny, strong margins.
Metals & Mining	TATASTEEL, JSWSTEEL, HINDALCO, COALINDIA	Capital intensive, cyclical operating margins, leverage-sensitive.
Power & Utilities	NTPC, POWERGRID, TATAPOWER, JSWENERGY	Regulated returns on equity, steady long-term dividend yields.
Capital Goods & Infra	LT, SIEMENS, ABB, BEL, BHEL	Long working capital cycles, order-book execution visibility.
Diversified Financials	BAJFINANCE, CHOLAFIN, SHRIRAMFIN, LICI	Consumer finance, lending spreads, underwriting asset quality.

Radar Chart Percentile Computation

Intra-peer percentiles are computed via SQL PERCENT_RANK() OVER (PARTITION BY peer_group_name, metric ORDER BY value). For inverse metrics such as debt_to_equity, percentile rank is inverted (1.0 - rank) so that higher values on the 8-axis radar chart universally indicate stronger financial health.

Chapter 6: Cash Flow Intelligence & Capital Allocation

Cash generation constitutes the ultimate test of economic earnings quality. Module src/analytics/cashflow_kpis.py evaluates cashflow statement dynamics across all companies, tracking CFO quality, CapEx intensity, free cash flow conversion, and 8-state capital allocation patterns.

Pattern (CFO, CFI, CFF)	Archetype Label	Corporate Capital Strategy & Interpretation
(+, -, -)	Shareholder Returns / Reinvestor	Healthy core operations generating surplus cash; funding CapEx and returning excess c
(+, +, -)	Liquidating Assets	Cash from operations bolstered by asset sales; used to repay creditors or fund distributi
(-, +, +)	Distress Signal	Operating cash flow is negative; business is kept afloat by selling capital assets and issu
(-, -, +)	Growth Funded by Debt	Aggressive early-stage expansion or heavy construction; CapEx and operating deficit fu
(+, +, +)	Cash Accumulator	Positive cash across all three activities; rare accumulation phase indicating capital buildu
(-, -, -)	Pre-Revenue / Cash Burn	Cash outflow across all activities; rapidly drawing down liquid treasury balances.
(+, -, +)	Mixed / Debt Expansion	Profitable operations supplemented by debt issuance for massive strategic expansions.
Unknown	Data Anomaly	Unclassifiable pattern due to missing cash flow reporting fields.

CFO Quality & CapEx Intensity Scores

- **CFO Quality Score:** Multi-year ratio of Cash from Operations to Profit After Tax (CFO / PAT). Scores > 1.0 indicate high quality accrual-free cash conversion; scores < 0.5 indicate earnings accrual risk.
- **CapEx Intensity:** Absolute investing outflow divided by revenue. Companies under 3% are classified as *Asset Light*; 3–8% as *Moderate*; and > 8% as *Capital Intensive*.

Chapter 7: NLP Text Extraction & Pros/Cons Heuristics

Financial disclosures contain valuable qualitative narrative text. The NLP subsystem parses unstructured analyst commentary from analysis.xlsx and algorithmically derives calibrated investment strengths (Pros) and concerns (Cons) using a 24-rule expert heuristic engine.

NLP Text Parser (src/nlp/parser.py)

The parser extracts historical growth rates and financial metrics using calibrated regular expressions matching text such as *'Compounded Sales Growth: 5 Years: 14%'*. Target fields include:

- **Compounded Sales & Profit Growth:** Parsed across 3Y, 5Y, and 10Y horizons and cross-validated against numerical CAGRs.
- **Return on Equity (ROE):** Extracted and verified against calculated equity returns.

Algorithmic Pros & Cons Generator (src/nlp/pros_cons_generator.py)

Rule Type	Signal Metric	Condition & Heuristic Threshold	Confidence Score
Pro Rule 01	Debt-to-Equity	Company is virtually debt-free (D/E < 0.05)	95% Confidence
Pro Rule 02	Interest Coverage	Strong debt service capacity (ICR > 8.0x)	90% Confidence
Pro Rule 03	Return on Equity	Good return track record (3Y Avg ROE > 20%)	90% Confidence
Pro Rule 04	Dividend Yield	Healthy dividend payout (Payout > 35%)	85% Confidence
Pro Rule 05	Working Capital	Working capital days reduced by > 10 days	80% Confidence
Con Rule 01	Interest Coverage	Low interest coverage ratio (ICR < 1.5x)	95% Confidence
Con Rule 02	Sales Growth	Poor sales growth over 5-year period (< 5%)	90% Confidence
Con Rule 03	Return on Equity	Subpar capital return (3Y Avg ROE < 10%)	85% Confidence
Con Rule 04	Valuation Multiple	Trading at elevated valuation multiple (P/B > 8.0)	80% Confidence
Con Rule 05	Earnings Quality	Tax rate divergence or high accrual gap	75% Confidence

Chapter 8: Machine Learning Stock Clustering & Archetypes

Module src/analytics/clustering.py groups all 92 companies into 5 distinct behavioral financial archetypes using KMeans clustering over Sprint 2 calculated ratios. The pipeline operates under strict data hygiene rules: non-financial identifiers are excluded, missing values are median-imputed, and features are standardized.

Cluster ID	Archetype Designation	Member Count	Financial Traits & Distinguishing Profile
Cluster 0	Growth-Oriented + High-Leverage	15 Companies (16.3%)	High 5Y revenue/PAT growth accompanied by above-average debt-
Cluster 1	Low-Leverage Companies	33 Companies (35.9%)	Conservative balance sheets, very low leverage, healthy operating r
Cluster 2	High Profitability Companies	2 Companies (2.2%)	Exceptional return on equity (>45%) and high operating profit margir
Cluster 3	High Profitability + Growth	1 Company (1.1%)	Fast-growth high-margin compounder exhibiting top-decile profitabili
Cluster 4	Balanced Financial Profile	41 Companies (44.6%)	Median market-average financial ratios representing broad stable blu

Clustering Validation & Diagnostics

- **Diagnostic Elbow Plot:** Tested across K = 1 to 10 inertia values; exported to reports/elbow_plot.png.
- **Correlation Heatmap:** Pearson correlation across the 10 clustering features exported to reports/correlation_heatmap.png to prevent collinear distortion.
- **Outlier Detection:** Identified 13 observations exhibiting |Z-score| > 3.0 across observed metrics, published to output/outlier_report.csv.

Chapter 9: Production REST API Reference & Concurrency

The backend API tier (src/api/main.py) exposes 16+ production REST endpoints under the /api prefix, fully documented with interactive OpenAPI Swagger UI at /docs.

Method	Endpoint Path	Description & Response Model
GET	/health	Health check: confirms API status, uptime, and database connection.
GET	/api/companies	List all 92 companies with ticker, name, sector, industry, and ISIN.
GET	/api/companies/{ticker}/profile	Detailed corporate profile with latest ratios, cluster, and pros/cons.
GET	/api/companies/{ticker}/pnl	Historical multi-year Profit & Loss statements.
GET	/api/companies/{ticker}/balance-sheet	Historical multi-year Balance Sheet statements.
GET	/api/companies/{ticker}/cashflow	Historical multi-year Cash Flow statements.
GET	/api/companies/{ticker}/ratios	Historical derived financial KPIs and quality scores.
GET	/api/companies/{ticker}/tearsheet	Binary stream download of the ReportLab 2-page PDF tearsheet.
GET	/api/screener/presets	List metadata and rule configurations for all 6 screener presets.
GET	/api/screener/presets/{name}	Execute named screener strategy and return matching company entities.
POST	/api/screener/run	Execute custom multi-metric filter queries.
GET	/api/sectors	Sector summary listing company count, average ROE, and margins.
GET	/api/sectors/{sector}	Detailed listing of member companies in a specific sector.
GET	/api/peers/{ticker}	Peer group comparison and 10 KPI percentile ranks for radar charts.
GET	/api/market-cap	Companies ranked by market capitalization with valuation flags.
GET	/api/portfolio/stats	Aggregate portfolio and index-wide performance metrics.
GET	/api/documents/{ticker}	Filing directory containing BSE annual report links.
GET	/api/clusters	KMeans cluster summaries, feature profiles, and company memberships.

Day 43 Performance Gate Benchmark Results

- **Concurrent API Load:** 10 concurrent requests to /health executed in **0.048 seconds**, well below the 10.0-second performance ceiling (100% success rate).
- **Port Conflict Verification:** Ports 8000 (API) and 8501 (Dashboard) verified free and conflict-free.

Chapter 10: Production Runbook & Acceptance Gates

This chapter provides standard operational procedures for executing pipelines, running services, and auditing production quality gates.

Step-by-Step Operator Runbook

1. **Environment Initialization:**
   uv sync or pip install -r requirements.txt
2. **Execute KMeans Clustering Pipeline:**
   python scripts/run_clustering.py (generates cluster labels, profiles, elbow & heatmap)
3. **Apply Database Query Optimization:**
   python scripts/optimize_sqlite.py (creates covering B-tree indexes)
4. **Verify Port Availability:**
   python scripts/check_ports.py (verifies ports 8000 and 8501)
5. **Launch REST API Server:**
   python src/api/main.py (listens at http://0.0.0.0:8000, docs at /docs)
6. **Launch Streamlit Analytical Dashboard:**
   streamlit run src/dashboard/app.py (access at http://localhost:8501)
7. **Run Full Automated Test Suite & Generate HTML Report:**
   python -m pytest --html=reports/pytest_report.html --self-contained-html

Acceptance Gates Verification Summary

All 20 acceptance gates defined in Sprint 6 have been verified and passed with zero critical failures. The formal signed checklist is preserved at docs/acceptance_checklist.pdf.

Role	Lead Name	Status	Date
Data Science & Analytics Lead	Priya Singh	VERIFIED & SIGNED OFF	October 2026
Backend API & Integration Lead	Vinay Todkar	VERIFIED & SIGNED OFF	October 2026
QA, Testing & Documentation Lead	Darshan B	VERIFIED & SIGNED OFF	October 2026
Project Sponsor / Team Lead	Lead Evaluator	APPROVED FOR PRODUCTION	October 2026